

Current Web, HTML, JavaScript, VBScript & ASP.NET

Lesson 63 — Markup, Scripting and Web History

JKSSB Computer Awareness

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Where This Lesson Sits

- ▶ A web page can be **static** (same content for every visitor) or **dynamic** (content built or changed at view time).
- ▶ **Markup** describes structure: HTML lays out headings, text, and links.
- ▶ **Scripting** adds behaviour: JavaScript and VBScript run small programs inside the page.
- ▶ **Server technology** such as ASP.NET builds pages on the server before they reach the browser.

The one idea

Markup structures the page; scripts make it act; the server can build it first.

HTML: What It Is

- ▶ **HTML** stands for HyperText Markup Language.
- ▶ It is a **markup language**, not a programming language: it tags text to say what each part is.
- ▶ An HTML document is an **ASCII-text document**: it can be written and read in any plain-text editor.
- ▶ The browser reads the tags and **renders** the page; the tags themselves are not shown.

Keep the pair straight

HTML describes structure; the browser decides appearance from those tags.

Tags and Attributes

- ▶ A **tag** names an element, written in angle brackets such as `<p>` and `</p>`.
- ▶ Most elements have an **opening tag**, content, and a **closing tag** with the slash.
- ▶ An **attribute** adds detail inside the opening tag, as `name="value"`, for example `<body bgcolor="yellow">`.
- ▶ A **valid** tag or attribute is one the language defines; an invented name is simply not recognised.

Worked example

In `<font size="4">Hello</font>`, `font` is the tag and `size="4"` is the attribute.

The h1 Heading

- ▶ Headings run **h1** through h6.
- ▶ **h1** is the **largest, top-level heading**; numbers grow as headings shrink.
- ▶ It is written as `<h1>Title</h1>`.
- ▶ Exam depth: h1 is biggest; h6 is smallest.

Trap alert

Do not reverse the order: **h1** is largest, not smallest.

The marquee Tag

- ▶ The **marquee** tag makes text **scroll** across the page.
- ▶ It is written as `<marquee>Welcome</marquee>`.
- ▶ It is a classic **presentation** tag: it moves text but adds no structure.
- ▶ It belongs to older, browser-specific HTML, not to strict modern structure.

One-line memory hook

marquee means moving text; h1 means the biggest heading.

DHTML: Dynamic HTML

- ▶ **DHTML** stands for Dynamic HyperText Markup Language.
- ▶ It is **not a separate language**: it is HTML plus style sheets plus scripts working together.
- ▶ It lets a page **change after loading**, for example text that reacts when the mouse passes over it.
- ▶ Contrast: plain HTML is **static**; DHTML pages are **interactive**.

Keep the pair straight

HTML lays out a fixed page; DHTML makes that page respond without reloading.

XML: What It Is

- ▶ **XML** stands for eXtensible Markup Language.
- ▶ It carries and **stores data**; it does not lay out a page the way HTML does.
- ▶ Tags are **user-defined**: the author invents tags such as <city> to label the data.
- ▶ Contrast: HTML uses a **fixed** tag set for display; XML uses an **open** tag set for data.

Trap alert

XML stores and transports data; it does not replace HTML for display.

Java vs JavaScript

Point	Java	JavaScript
Type	Full programming language	Page scripting language
Runs	Compiled to bytecode	Interpreted in the browser
Needs	Java platform	Just a browser

Trap alert

Similar names, different languages: **Java is not JavaScript.**

JavaScript: History in One Line

- ▶ **JavaScript** is a lightweight scripting language for web pages.
- ▶ It was developed at **Netscape** for the Navigator browser in the mid-1990s.
- ▶ It first shipped under other names (Mocha, then LiveScript) before the **JavaScript** name was settled.
- ▶ Despite the name, it was **not** a cut-down Java; the name was a product of that era.

Exam anchor

JavaScript means Netscape; VBScript means Microsoft.

JavaScript String Concatenation

- ▶ In JavaScript, the `+` operator **joins (concatenates)** strings.
- ▶ `"Good" + "Morning"` gives `"GoodMorning"`.
- ▶ Mixing text and a number with `+` joins as text: `"Total: " + 5` gives `"Total: 5"`.
- ▶ The same symbol adds numbers, but with strings it joins them.

Worked example

`"Web" + "Page"` evaluates to `"WebPage"` — joined, not added.

VBScript and Dim

- ▶ **VBScript** stands for Visual Basic Scripting Edition, from Microsoft.
- ▶ It scripts pages and small Windows tasks in a Basic-like style.
- ▶ Variables are declared with the **Dim** statement, short for Dimension.
- ▶ `Dim city` declares a variable named `city`.

Exam anchor

See `Dim` and think VBScript variable declaration.

ASP.NET and the Session Object

- ▶ **ASP.NET** is Microsoft's framework for building dynamic web pages on the **server**.
- ▶ The server builds the page and sends plain HTML to the browser.
- ▶ The **Session** object stores data for **one user across many pages**, for example a login name.
- ▶ Contrast: Session is per-user; Application state is shared by all users.

Trap alert

Session keeps one visitor's data; it is not shared across visitors.

Meta-Search Engines: Dogpile

- ▶ A **search engine** keeps its own index; a **meta-search engine** keeps none and queries other engines.
- ▶ **Dogpile** is the classic exam example of a meta-search engine.
- ▶ It collects results from several search engines and shows them together.
- ▶ Google and Bing are search engines; Dogpile searches the search engines.

One-line memory hook

Dogpile piles up other engines' results.

Valid Tags and Attributes: Drill

- ▶ Valid tags include `h1`, `p`, `marquee`, `font`, `body`, and `html`.
- ▶ Valid attributes sit inside the opening tag as `name="value"`, for example `size`, `bgcolor`.
- ▶ A tag and an attribute are **different things**: `font` is a tag; `size` is its attribute.
- ▶ Case in the stem does not change the answer: match the **name**, not the capitalisation.

Trap Alert: Facts to Keep Straight

- ▶ HTML is an **ASCII-text** document, not a compiled binary.
- ▶ **h1** is largest; **marquee** scrolls text.
- ▶ **Java** is not JavaScript; **XML** carries data, HTML displays it.
- ▶ **+** joins strings in JavaScript; **Dim** declares in VBScript.
- ▶ **Session** is per-user; **Dogpile** is meta-search.

Lesson 63: Recall

- ▶ HTML (HyperText Markup Language) is a markup language stored as an ASCII-text document; tags structure, attributes refine.
- ▶ h1 is the largest heading; marquee scrolls text.
- ▶ DHTML is HTML plus styles plus scripts for pages that change after loading; XML is extensible markup for carrying data.
- ▶ Java is a full compiled language; JavaScript is a Netscape browser scripting language, and + concatenates its strings.
- ▶ VBScript declares variables with Dim; ASP.NET keeps one user's data in the Session object.
- ▶ Dogpile is the classic meta-search engine: it combines other engines' results.